

CSE 291A Final Project FALL 2025

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Emotion Recognition for Autistic People

Section 1: Summary

Autism Spectrum Disorder (ASD) compromises individuals' abilities to recognize emotions, posing an obstacle for them when this skill is necessary. I initially suggested an AI facial expression classifier as the solution to this problem, but found it lacking. Specifically, it does not improve long-term understanding for users and artificially constrains the problem to social interactions. To improve on this shortcoming, I designed a tool that includes both teaching and multi-modal classification, leveraging AI's advantages in accessibility and adaptability. The user can use their AI tool in a multitude of situations, have multi-modal input, and the tool adapts to their use case.

First, the tool automatically generates lesson plans for the user based on what they need help with. For instance, it teaches them what signs to look for to classify emotions, how to approach others for clarifying questions, and works through practice examples with them. This method strengthens the user's long-term understanding and communication skills. Second, users can input what they have difficulty with in text, on-screen (e.g., in a movie), or in real life through words or pictures, and receive emotion recognition. The model will also provide a confidence score, indicating how confident it is in its prediction. This multi-modality solves a previous iteration's problem of relying on photos of people, although this option remains available. If obtaining consent to do so is difficult, the user can explain the situation to the model through words instead, protecting stakeholders from social, privacy, and legal risks.

The HCAI tools I used include Storyboarding, Framing Template, Stimulated Recall, Prototyping, Wizard of Oz Methodology, Reward Function Template, Success Metrics Framework, User Needs Mapping Template, Questions for Domain Experts, AI Fairness 360 (IBM), Arize Bias Tracing, Model Cards, Partial Dependence Plots, Card Sorting, A/B Testing, Versioning, and System Usability Scale.

Section 2: Related Work

Levy et al. (2024) made an environmental translator using small vibrations and color-coded feedback to interpret and convey emotional states for people with ASD. Their results show that adults with ASD could adapt to their system and immediately recognize others' emotions. However, the study was conducted in a controlled laboratory setting and with 10 participants, making it difficult to generalize its results to an outside setting and other individuals. Additionally, their hardware consists of vibration motors attached to smart glass, which the paper does not view as a limitation but instead claims provides a pleasant user experience. Since this claim is not supported with evidence, I believe that further testing needs to be conducted to ensure it does not induce societal risks, such as interfering with users' daily lives or making them feel alienated. Despite its shortcomings, this study supports the idea that emotion recognition can be teachable. Learning from their limitations, my app attempts to be less invasive, in particular by using multimodality to avoid societal risks.

Alternatively, Chiti et al. (2025) present an emotion understanding game that acts as a therapy tool, addressing the question of how to teach emotion recognition over time. They found that multi-modal feedback, particularly music and color, enhances emotion recognition. Their success metric focused on the platform's usability and was measured through the System Usability Scale (SUS) and qualitative feedback from therapists. Specifically, high scores on the SUS and positive feedback from therapists showed that their platform has potential as a tool. However, they reported that the small sample size, short intervention duration, and lack of a control group require future studies with larger participant groups to validate their findings and assess long-term effects. Similar to the previous study, this work validates the learning approach for emotion recognition present in my app.

Addressing algorithmic bias mitigation in facial expression recognizers, Galán et al. (2023) explored using a debiasing variational autoencoder. They found that their method improves the model's accuracy on categories that are underrepresented without explicitly identifying those categories beforehand. Despite their success, they discuss the continued existence of a disparity between categories that requires additional bias mitigation strategies. Similarly, Ge and Chen (2025) examine bias in emotion recognition models, but across multi-modal input. They argue for a modality weighting strategy to improve multi-modal models. While they present an effective solution, they acknowledge that it is architecturally simple and requires further research for highly noisy conditions. Both of these papers introduce theoretical and technical solutions that could be applied to mitigate bias in my own emotion classifier.

Section 3: Define

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder that compromises individuals' abilities to recognize emotions, posing an obstacle for them when this skill is necessary. In developing this problem statement, I was going to presume the following: in social interactions, people with ASD may either misunderstand or be misunderstood by the person they are communicating with, thus experiencing an obstacle in their daily lives. This sentence is a pitfall due to its assuming the behavior of the user (presuming) and artificially constraining the problem (hamstringing). It is true that those with ASD have an alternate ability, but their daily experience cannot be generalized, and the solution does not necessarily need to rely on socially-interactive contexts.

Different approaches to this problem include helping individuals with ASD recognize emotions in real-time, providing them with a therapy tool, and helping their parents understand them (Levy et al., 2024; Talaat, 2023; Chiti et al., 2025). Additionally, systematic literature reviews point out that studies commonly focus on facial expressions, prosody of speech, and physiological signals, and target autism in a broad sense rather than focusing on specific parts of the spectrum, seldom addressing privacy or security issues (Landowska et al., 2022; Banos et al., 2024).

The storyboard in [Figure 1](#) is developed from the angle that their compromised emotion recognition makes individuals with ASD's social interactions difficult, as they struggle to identify others' emotions. Here, the stakeholders are the conversation participants. The AI acts as a translator or classifier for the individual with ASD so they can recognize the other participant's emotions. This technology may take the form of an app that uses the phone camera, which the individual with ASD would point at the other participant.

Always-on topics, such as legal risks, human safety, societal risks, and privacy and security, are especially involved when it comes to the discrepancies between how I expect the system to be used compared to how it might actually be used. For instance, I expect it to be used with the consent of the other party and for the sole purpose of emotion recognition. Using it without the other party's consent or knowledge can warrant being sued, and verge on harassment. It is simply invasive. Additionally, it may stigmatize the user if they broach social norms to use the app. Furthermore, the app should not save the image of the other party's face and collect data, but bad actors could potentially take over the system and do so. This action may be through the user, such as by receiving consent for using the app but then using their phone's screenshot feature, or by a hacker. I received feedback that requiring obtaining consent may make usage of the app more difficult, as some users may feel intimidated to ask strangers for permission to use the app. This issue will be addressed in the design of the app.

Using the framing template in [Figure 2](#), I began with the definition that compromised emotion recognition affects people with ASD and those they interact with.

Looking outside the frame, the problem is not contained to social interactions. Rather, it could involve experiences such as watching a movie, reading, introspection, and self-regulation. Looking in the mirror, the focus on AI in the original solution contributes to the problem by being a quick fix and not developing personal understanding. On reflection, a better goal than classifying emotions for them is teaching them how to perform the classification on their own. Examining bright spots suggests that users across the autism spectrum will have different levels of understanding and their own methods of emotion recognition. Additionally, some users may not have this problem because they regularly communicate what they do not understand. Then, the solution should be applicable to various parts of the spectrum, and communication skills can be incorporated into the solution. Taking the user's perspective, they are hindered by a lack of information that arises from neurotypical assumptions. Moving forward, I alter the solution to be a long-term fix, focusing on equipping the user with tools to not only understand others, but understand themselves as well.

AI's unique value here is its accessibility and adaptability; it can be used in a multitude of situations, process multi-modal input, and adapt to custom use cases at any time. It solves the problem of not knowing how to identify emotions, which is otherwise difficult for a person with ASD to do. The AI tool can teach the individual what signs to look for to classify emotions, assist with pieces of text, and identify facial expressions through pictures. None of these abilities are unique to AI, but the combination of their multi-modality is. Alternative approaches could include having a human teacher or self-studying through flashcards. However, having someone else present at all times is unrealistic, and self-studying tools cannot process new input or adapt to the user's level. Thus, AI is the correct approach for this problem due to its ability to work with multi-modal information, adapt to the user, and remain readily accessible.

The part of the system that will be automated is the generation of lesson plans for the user. Based on what the user needs help with and how much help they need, the lessons will automatically adapt to them, making automation a helpful choice. The degree of delegation for this task will be AI-led and human-assisted. Human developers provide the information for the AI and the user gives it feedback, allowing control over the input and personalization over the output. Augmentation will play a role in assisting the user with real-time input. This enhancement will allow them to recognize emotions that they are having difficulty with, in-text, on-screen, or in real life. Its degree of delegation will be human-led and AI-assisted, ensuring the user has control over privacy settings, assuming they know when and what is appropriate to record. The AI only provides outputs for what the user specifically inputted.

Figure 1

Storyboard

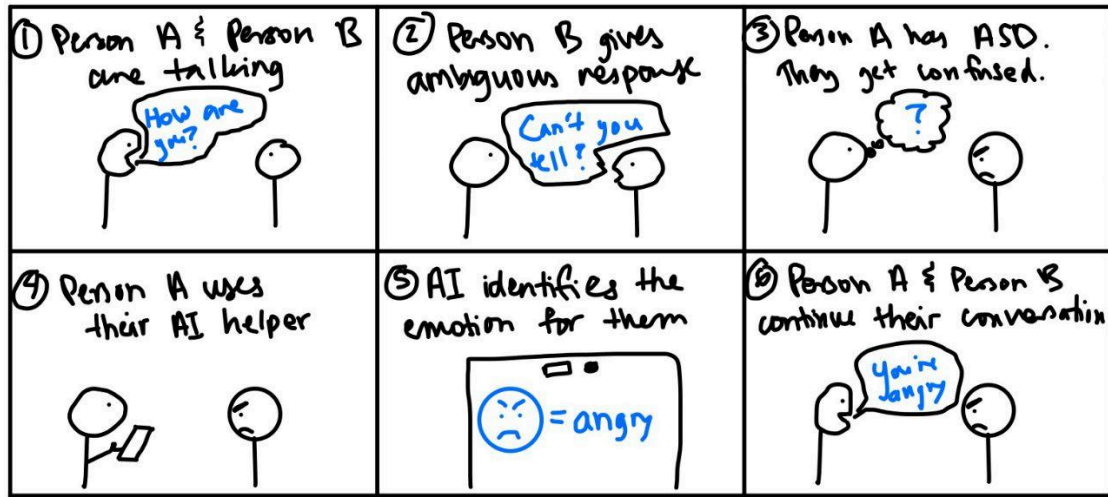
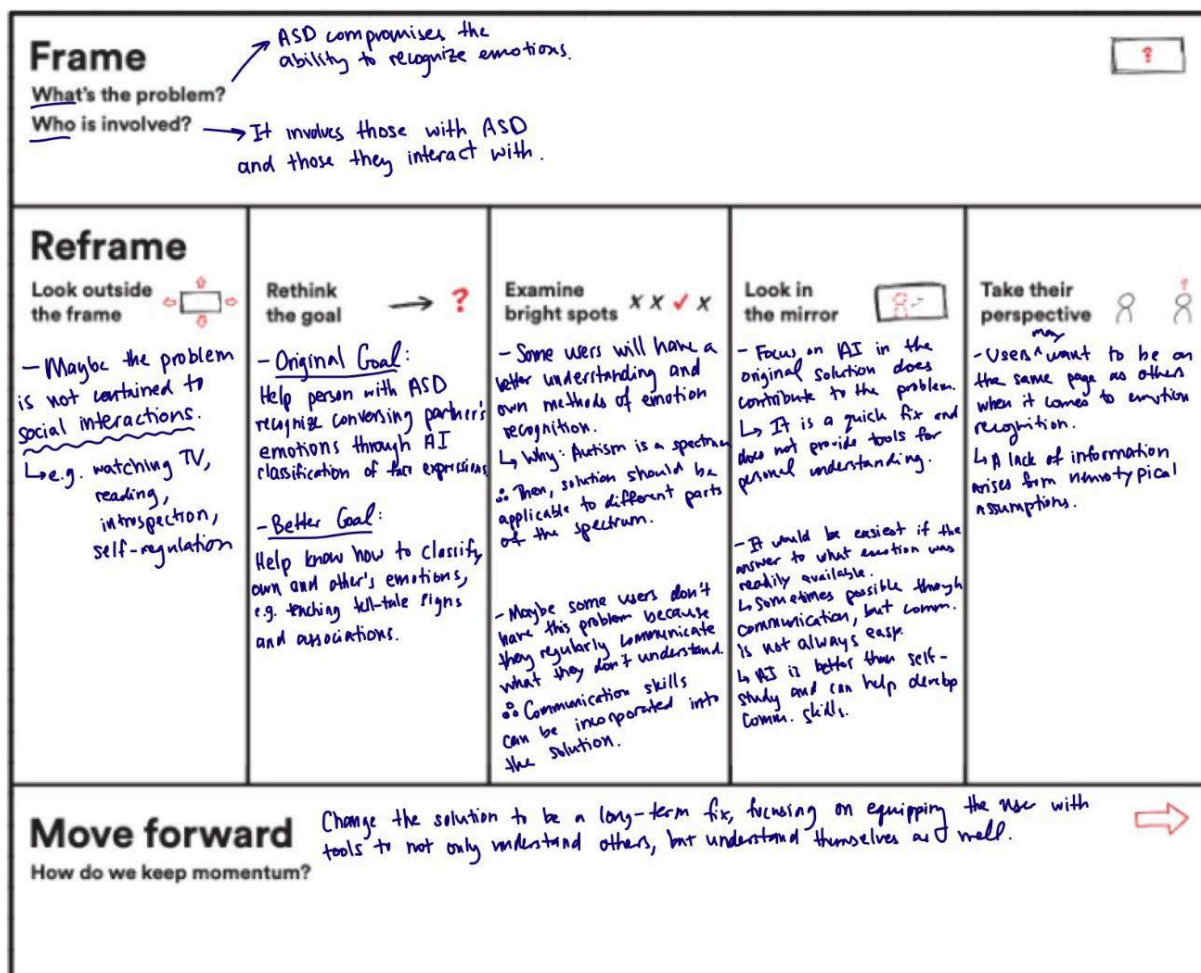


Figure 2
Framing Template



Section 4: Design

I would use stimulated recall instead of the think-aloud protocol for this study. Since the tool targets an area of difficulty, I worry about users being embarrassed to voice their thoughts or use certain features during the think-aloud protocol. I do not want them to feel like they are being judged, as that would interfere with discovering the suitability of the design. Instead, I want to see how they interact with the features without disruption, and afterwards clarify their thought process. Through this method, I could then identify how to improve the design to better suit their needs.

I created a prototype of the app's user interface, illustrated in [Figure 3](#), to identify the features. There are mainly two categories for the user to interact with: Learn and Classify. Learning entails participating in an AI-curated personalized course, reviewing topics, accessing tips, and practicing with examples. This category enables long-term understanding of methods for emotion recognition and supplements self-conducted classification. Classify leads the user to a screen where they can choose between text and image mediums. Text leads to an LLM model, where the user can explain context, ask questions, or insert excerpts to receive an emotion classification. This option is also useful when it is difficult for users to obtain photos, addressing the previous feedback I received. Alternatively, in Image, the user may upload a photo for an AI model to recognize the emotion of. If the photo was taken personally, the user must confirm that they obtained consent. This process would protect the app developers from lawsuits and remind the user of people's privacy rights.

I would use a Wizard of Oz system for early prototyping because it would be cheaper to simulate the AI models' responses rather than spend time developing models that will be tweaked with every design update. In the worst-case scenario, the developed model may not even be used. Through the Wizard of Oz system, I could find out what UI users prefer, which features they like and dislike, and what additional features they want. Theoretically, I would allow the participant to press the different buttons in the app, and then type what the AI's answers may be in response.

Participatory design would be helpful in ironing out the system, especially because ASD is a spectrum, and I want to serve people with diverse needs across the spectrum. Involving them in the process would also allow me to hear what they really need instead of making assumptions. It would help remove the bias I bring to the project and make the tool more inclusive.

I filled out the reward function in [Figure 4](#) for the lesson plan generator and emotion classifier, since these are the AI-specific features of my tool. For the lesson plan generator, the AI model will be optimized for precision so the user does not receive unsuitable or untimely lessons. The tradeoff for choosing this method means the model will sometimes skip over lessons that are appropriate or that the user wants to see. However, it is better for them to not see what would not serve them. Meanwhile, the emotion classifier will be optimized for recall because it is in the user's best interest to

recognize emotions that are present, for example, in social interactions where it is crucial to infer that the other person is angry. The tradeoff for this method is that the model will sometimes predict emotions that are not present. The hope is that this misclassification can be cleared up through communication and further attempts.

Explanations of both AI tools are needed for the users so they understand how to use the system, how much they can rely on it, and how protected they are. I would explain that the lesson plan adapts to them as they use it more often while keeping their data private and secure. Instead of the algorithmic details, I would state what it uses to make predictions, such as the frequency of correct answers, the time spent reading a lesson, or revisited topics. For the classifier, I would explain that the system is making its best guess, but is still able to make mistakes, such as predicting emotions that are not present or missing emotions that are. Other stakeholders, such as investors, need an explanation for what measures the system takes to protect privacy and against financial risks. I would explain what data the system collects, what is encrypted, and how it delegates the responsibility of obtaining consent to the users.

Users should trust that the system is capable of creating lesson plans for them that will help their emotion recognition and communication skills. They can also trust that the emotion classifier will be correct most of the time, but be aware that it can occasionally make mistakes. I received peer feedback to be more concrete about how I plan to achieve this trust. Therefore, I will do so by explaining the system to users and being upfront about its limitations. I will also open-source the models and communicate how the system takes action on user feedback.

In terms of control, users can provide feedback on lesson plans, prompting the system to adapt to what they like or dislike. I will have a button available on each page of a lesson plan that they can optionally thumbs up or down. Additionally, I will have a page they can navigate to if they want to type out their feedback. For the classifier, what the system makes predictions on is completely up to what the user chooses to input. The input will be in the form of a text box for the user to type in or a place for them to upload a photo. The classifier will not make predictions unless prompted to do so.

I filled out the success metrics framework in [Figure 5](#), focusing on the image and text classifiers being equally accurate and accessible. I arrived at the final version: *If the ratio of the text classification to image classification usage goes above 70%, we will conduct user studies to investigate the difference between their experiences with the two classifiers.* It is a success if both are explored on day 1 and continue to be used in the long run. To address success metric failure, we will focus more on understanding user experiences than trying to match the exact threshold. It is okay if users prefer one model; the goal is to make sure they are being properly served. I received feedback to consider additional success metrics. These, then, can include users completing more than 90% of lesson plans, pressing thumbs-down on a recommendation less than 15% of the time, and returning to the app at least once a week.

Figure 3
Prototype

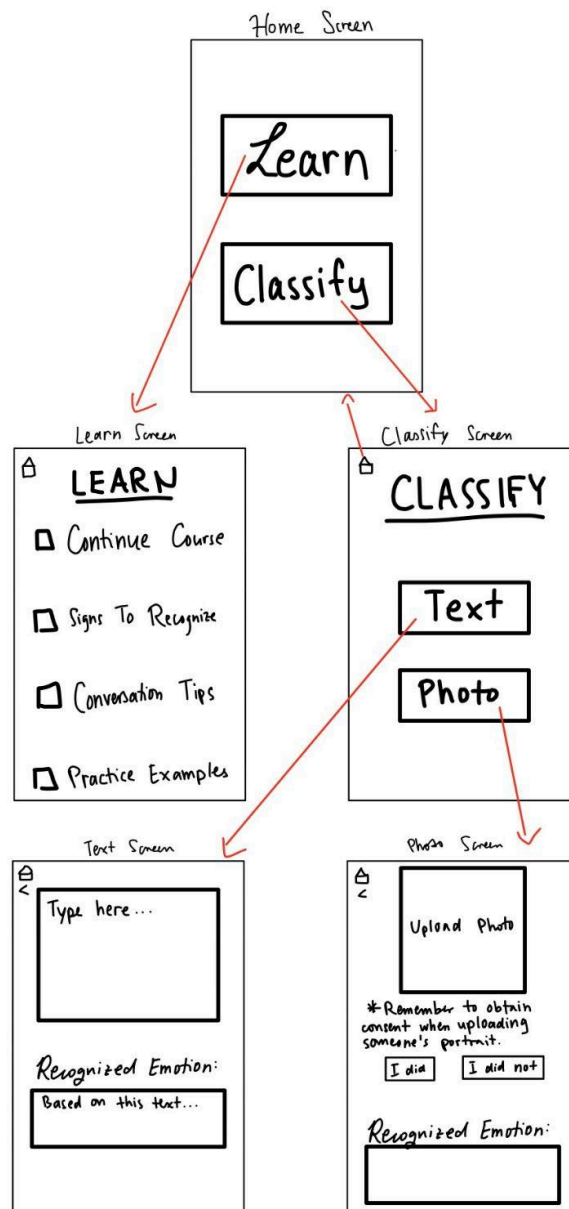


Figure 4
Reward Function Template

	Positive Prediction	Negative Prediction
Positive Reference	{Course materials: the chosen lesson is at the correct level for the user}	{Course materials: the model judged a lesson was not at the correct level for

	{Course materials: the chosen lesson is information the user wants to learn presently.} {Classifier: the model predicted the correct emotion.}	the user and did not integrate into the lesson when it really was at their level.} {Course materials: the model judged a lesson was not timely when the user actually wanted to learn it presently.} {Classifier: the model failed to detect the correct emotion.}
Negative Reference	{Course materials: the model judged a lesson at the correct level for the user and when it was not really suitable.} {Course materials: the model judged a lesson was timely when the user did not want to learn it presently.} {Classifier: the model predicted an emotion that was not actually present.}	{Course materials: the model judged a lesson was not at the correct level for the user and it really was not at their level.} {Course materials: the model judged a lesson was not timely when the user really did not want to learn it presently.} {Classifier: the model correctly identified that a given emotion is not present.}

Figure 5
Success Metrics Framework

If __{ specific success metric }__
for __ { your team's specific AI driven feature }__
{ drops below/goes above }__ { meaningful threshold }__
we will __{ take a specific action }__.

Version 1

If users favor text classification over image classification more than 80% of the time, we will conduct user studies to understand why they are not using the image classifier.

Version 2

If users favor text classification over image classification more than 70% of the time after a week of usage, we will conduct user studies to investigate if there are issues with the image classifier.

Take each version through this checklist:

- ☐ Is this metric meaningful for all of our users?
 - ☐ How might this metric negatively impact some of our users?
- ☐ Is this what success means for our feature on day 1?
 - ☐ What about day 1,000?

Final version

If the ratio of the text classification to image usage goes above 70%, we will conduct user studies to investigate the difference between their experiences with the two classifiers.

Section 5: Data

I should consult emotion researchers and clinical psychologists for the content of the lesson plans, and machine learning engineers for the AI generator and classifier. Referencing the Questions for Domain Experts tool, I would ask the following.

(1) Important data

- (a) **To the clinical psychologists:** “When designing tools for teaching emotional skills, what kinds of client information or behavioral data are meaningful? What makes this data reliable or unreliable?”
- (b) **To the ML engineers:** “What features improve customized lesson plan generation or emotion classification?”

(2) Collection

(a) Clinical psychologists: “How is emotional data captured in practice? Are there trusted sources or validated instruments you recommend when collecting or annotating emotional/behavioral data?”

(b) ML engineers: “How are datasets for instructional systems usually curated? What’s the standard practice for annotating emotion-recognition data for classification tasks?”

(3) Problems

(a) Clinical psychologists: “What issues commonly arise with emotion/communication data (bias, missing details, labeling)?”

(b) ML engineers: “What data-quality issues tend to break emotion classifiers or generation models (cultural variability, low-quality images)?”

(4) Impactful circumstances

(a) Clinical psychologists: “Do situational factors (e.g., setting, rapport, stress level, time of day) meaningfully affect emotional expression or communication during assessment?”

(b) ML engineers: “Are there temporal or behavioral factors that make the underlying data hard to model for lesson-plan generation? Are there contextual variables that can reduce the classifier model's robustness?”

(5) Reusability

(a) Clinical psychologists: “How feasible is it to reuse clinical emotional-behavioral data outside the original therapeutic context?”

(b) ML engineers: “Is it easy to repurpose existing education-focused datasets for generative lesson planning or emotion datasets for real-world use cases?”

(6) Awareness

(a) Clinical psychologists: “What should a product team be most cautious about when working with emotional data from real clients?”

(b) ML engineers: “What are the key engineering considerations when working with pedagogical content data or deploying emotion classifiers?”

The lesson generator leverages features that help it decide what to teach and how to sequence content. These include the frequency of completed lessons, what topics are revisited, the time it takes to complete lessons, and user accuracy scores. This information allows the model to adjust its weights to be more personalized to the end user’s preferences and level, giving them a better experience with the AI tool. As a result of this choice, security measures would have to be put in place to protect the user’s privacy and not tie their data to them from the perspective of data collectors.

For the emotion classifier, features include sentiment cues and contextual embeddings for text input, and facial expressions, body posture, and other contextual cues for visual input. Using more specific features would improve the model’s transparency and decrease the potential for bias. Allowing the model to extrapolate its

own features improves classification abilities, but sacrifices those aspects. The end user would benefit from a model with high accuracy, but not one that suffers from bias. More broadly, misclassifications pose a social risk for users who utilize it in social settings, causing misunderstandings between them and others.

I filled out the table in [Figure 6](#) based on this summary of my tool: *Building an emotion recognition classifier and a lesson plan generator to improve understanding of emotion recognition and communication skills*. I used the features I have previously listed and thought of how to translate my goal to more tangible datasets. I also wanted to retain some interpretability and account for diversity.

I would use existing data for the classifier. Since emotion classification is a widely worked on task, there is already a large amount of existing data that can lead to models with high accuracy. However, I would procure data myself for the lesson generator. My goal is to create a curriculum that is helpful for people across the autism spectrum. Thus, I would consult research on teachings that have been helpful and group them by topics and levels of difficulty to procure the dataset.

To see if my classifier is biased, I would check if there is a pattern to its misclassifications, such as by examining its accuracy across demographics, picture quality, patterns of speech, and emotional expressions. For the generator, I would look at the personalization fairness, seeing if some users are always given a specific type of lesson or level of difficulty, and whether the model accounts for growth. Additionally, I would examine the datasets to make sure the demographics are representative, labels are evenly distributed, and there is high annotation agreement.

Sensitive information could include personal details the user entered in text, uploaded photographs of people, or user performance scores related to communication and emotion recognition skills. Procuring and using this data has identification and privacy risks, consent and legal risks for photographs, and societal misuse risks. While these risks must be taken seriously, proper security measures could be put into place to mitigate them. With mitigation strategies in place, procuring this data is worth the residual risk since it allows the tool to give useful and accurate responses.

Figure 6
User Needs Mapping Template

Type of Data	Scope of Data
Users	Individuals with ASD
User action (core value prop)	Recognize emotions through skill development or a classifier.
ML system output	Lesson plan generations, emotion classifications.

ML system learning	Types of lesson plans the user needs/wants, emotion classification task
Training dataset needed	Sequential data used to teach emotion recognition, text/image-to-emotion data
Key features needed in dataset	• Completed lessons • Return rate to a topic • Duration of lessons • User accuracy scores • Textual embeddings • Image embeddings
Key labels needed in dataset	• User likes/dislikes output • User's feedback for rejections
Data formatting	• lesson_id (lowercase string) • user_feedback (lowercase string) • text_input (string) • image_input (array) • emotion_label (lowercase string)
Real world data considerations	Does the dataset... • Account for people across the spectrum • Account for diverse bodies of text and images • Reflect the user's cultural/social settings
Data source key user questions	"How does the app know what lesson to show me next?" "How does the classifier make its predictions?"

Section 6: Build

My tool requires architectural decisions for a lesson generator and text/image classifiers. Deep learning is not necessary for generating lesson plans; the task can be completed with reinforcement learning (RL) based on user feedback. I would build an initial model using rule-based methods on the features and then update its parameters using RL. The greatest strength of this architecture would be its interpretability and not needing as much data. However, it sacrifices the greater personalization skills of a deep learning model, requires more time for a developer to outline rules, and may lose out on

great predictive features that the developer is unaware of. Despite these shortcomings, the lesson plan generator will not be a complex tool and does not require much more than rules based on the user's level and what they've already covered.

On the other hand, I would implement a deep learning architecture for the classification model. Deep learning models are highly skilled at extracting emotions from text and images, being able to support what a person with ASD struggles with. Using this architecture sacrifices interpretability and requires more data, but makes up for it with greater performance. I would also use a pretrained classifier for the sake of sustainability, as there are already well-performing emotion recognition models out there. Such models may be susceptible to biases, such as weighting different dialects or demographics unevenly, which I will mitigate by researching the models carefully before applying them and fine-tuning them if necessary.

To pre-process the data for lesson generation, it is important to note that it is a sequential model, depending on what has already been generated and how the user interacted with it. Therefore, I would need to sort it chronologically, label the lessons to identify their level of difficulty and the topic they cover, and also categorize the user history by completion status of the lesson, performance, and feedback rating. Since it is time-dependent, I would split the data temporally. I wouldn't use dimensionality reduction because it would lose important data needed to make predictions.

For text classification, I would clean the text and apply a tokenizer to feed into the deep learning model. Image classification may require preprocessing such as aligning the faces and adjusting the image size and pixel values. I would split the data randomly while preserving the emotion distribution so the model would not be skewed. Since deep learning models already perform dimensionality reduction internally, I would not do so myself to avoid the risk of losing important information.

As previously mentioned, the lesson generation requires features that convey temporal order, information about the lesson, and information about the user in order to present lessons that are appropriate and engaging for the user. Such features would include lesson difficulty level and topic, user level, lesson completion history, and prior user feedback. The classification models do not need in-depth feature extraction, since deep learning can learn associations itself. Such feature selection could slow down its training and use more computing resources. If it is found to increase model accuracy, to pass on domain-specific knowledge, features could include text semantics and contextual information.

Both models require feedback in the training process. The lesson generator is based on RL; users' liking/disliking the lesson, completion, and accuracy would be the reward function that improves the model's personalization. The classifier would also incorporate feedback about user likes, amount of use, and accuracy to improve the model predictions.

Algorithmic bias could take the form of aggregation bias. In the lesson plan generator, it could show up by recommending the same lessons to individuals regardless of what it is they find difficulty with and the level to which they struggle with it. As for the classifier, it could show up by not accounting for age, regional dialect, or demographic diversity. I need to combat this bias to make my tool accessible across the ASD spectrum and generalizable across populations.

Another form of model bias is evaluation bias. For the lesson plan generator, benchmarks may be based on users who prefer lessons in a certain sequence. Individuals who prefer to focus on one topic before moving on to the next, or conversely, individuals who prefer jumping around between topics, could be left out. In the text classifier, benchmarks could be based on a specific regional dialect or type of setting. Then, different speakers and different levels of formality would be miscategorized. The facial classifier could miscategorize images that are angled or lighted differently from the benchmark. To improve the model's accuracy across a range of settings, this bias must be combatted too.

I could use the AI Fairness 360 and Arize Bias Tracing tools for help identifying model biases. The former examines the datasets that the models are trained on, while the latter looks for features and cohorts that may be contributing to algorithmic bias. Through identifying the problem with these toolkits, I could then retrain the model to mitigate its bias.

Section 7: Test

The lesson plan generator will be transparent in how it makes decisions because it is rule-based. It should also be transparent in how the feedback adjusts its weights during RL, which could be accomplished by using interpretable features such as lesson difficulty level and topic, user level, lesson completion history, and the user's thumbs-up/down response. It can also have inspectable settings so that a user can see what the system believes about their learning style. Meanwhile, the emotion classifier should be transparent in what features it is using to make decisions in order to ensure it is not being biased towards certain dialects, demographics, or anything that can cause incorrect or harmful responses. To accomplish this, interpretability tools can be used to verify that the model is not leaning on biased markers or making unwanted associations. Additionally, for both cases, the model card in [Figure 7](#) will help increase transparency by providing insight into the models' components.

The model should be explainable enough that end-users can understand what the model is basing its decisions on and how their data is being used, because misinterpretation could harm user trust. However, they do not need to know the detailed logistics and calculations the model is making. It is sometimes enough to know how

much the features will influence the decision. For instance, Partial Dependence Plots (PDP) could be helpful to understand the lesson plan generator, so the user will know that how much time they spend on a lesson, how much they complete it, and what feedback they provide will adjust the personalization. PDP will be less helpful for the classifier, since the end-user does not need to know how the deep learning model is learning. It is more useful to them to know how well the model performs, which can be shown to them through confidence indicators.

The UI is illustrated in [Figure 8](#). It is an adaptation of [Figure 3](#), but with the additions of “Model Info” and “Give Feedback” on the Home Screen and a Lesson Screen. “Model Info” shows inspectable settings, interpretability tools such as the PDP, and the model cards. “Give Feedback” allows the user to input textual feedback about the app. As before, there are two main features: “Learn” and “Classify”. “Learn” entails participating in a personalized course that the model curated, reviewing topics, accessing tips, and practicing with examples. This category enables long-term understanding of methods for emotion recognition and supplements classification that the user wants to do on their own. When accessing the course, the user has the option to indicate whether they like or dislike the lesson. “Classify” leads the user to a screen where they can choose between text and image mediums. “Text” leads to a place for the user to explain context, ask questions, or insert excerpts and receive an emotion classification from the model. Alternatively, in “Image”, the user may upload a photo for the model to recognize the emotion of. If the photo was taken personally, the user must confirm that they obtained consent, protecting against legal and privacy risks.

To match users’ mental model to the UI, I would try to understand what they expect when using the app in terms of how to navigate and what information they look for. Card sorting would be useful in seeing how they categorize the app’s features and ideas. As mentioned as part of the HCAI tool, I would write the names of the different app features with their description on index cards and give them to users to sort into piles. Features would include the learning aspects of the model, such as the lessons, tips, and practice problems, the text and image classification aspects, and places for user feedback.

Personalization differs for the two models. On one hand, the emotion classifier does not need to be personalized. Doing so would needlessly collect user data without substantially improving model accuracy. On the other hand, the lesson plans need to be personalized. It would depend on data such as when users viewed previous lessons, how much of the previous lessons they completed, the lessons’ associated level of difficulty and topic area, and the user’s inferred level of learning and topic interest. To procure this data ethically, I would inform the user ahead of time that these variables will be monitored but not tied to them in databases. Information will either only be present at their specific instance or anonymized if collected for a dataset. I would obtain their

consent, allow them to toggle the option on and off, and show them how the lessons would be generated with and without the data collection.

Both models have limitations. The lessons are based on information that would be subject to whenever the database was last updated. Additionally, when the plans are generated, they are done so with preset rules and may miss nuances in learning styles. I would include a notice in the UI explaining these limitations so users know when the information was last updated and not to have unrealistic expectations for the lesson plans. The emotion classifier is subject to mistakes and can struggle with ambiguous or noisy input. To communicate its limitations, I would include a confidence indicator, showing how confident the model was in its prediction, and use clear wording that communicates its output is just an estimate and should not be taken as fact.

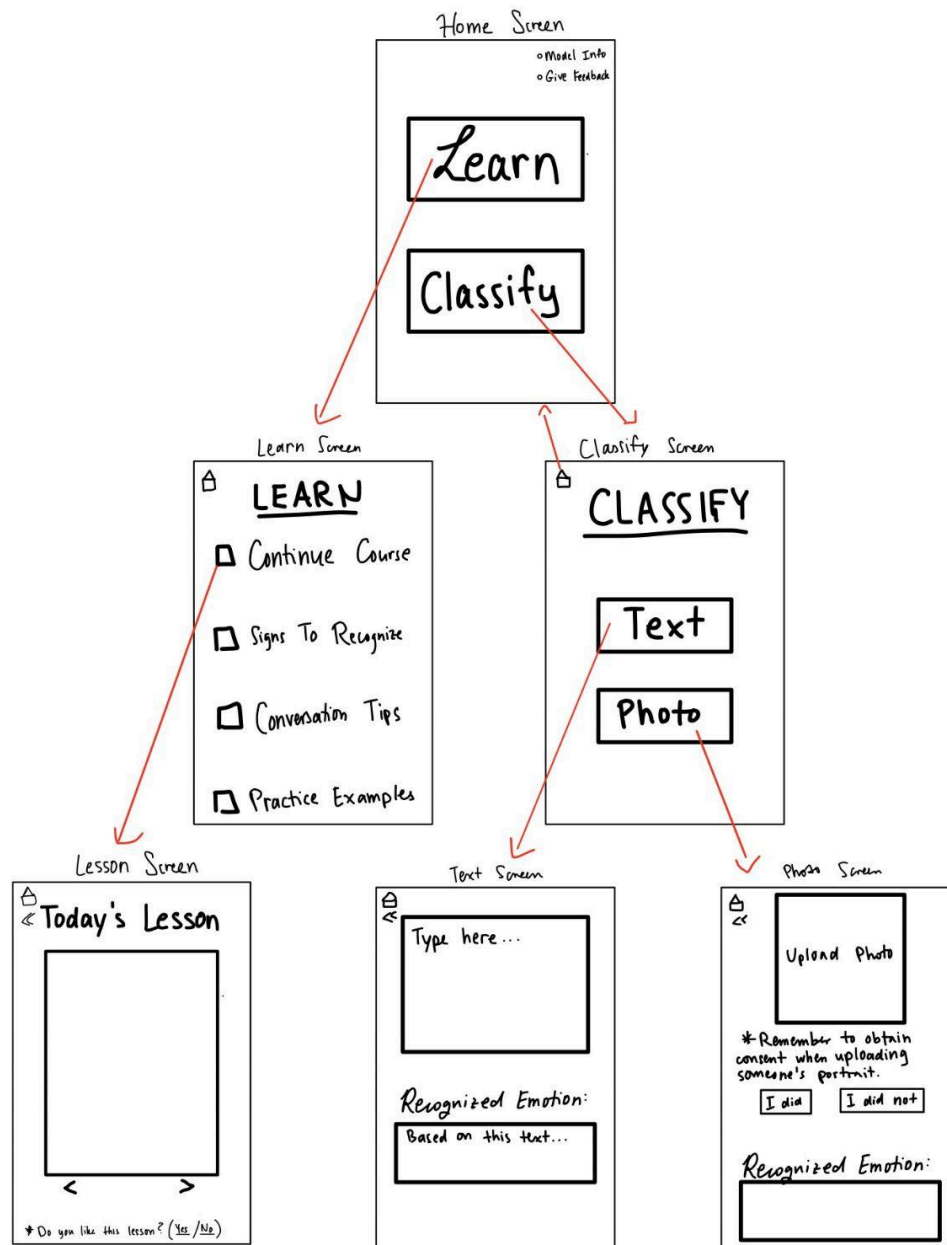
Iteration would allow the system to better align with user needs and is thus necessary. I would use it to update the model based on the aforementioned success metrics, monitoring user feedback and history, and ensuring model fairness. A/B Testing would be helpful to find the interface that best serves the user and aligns with their expectations for the system. To do so, I would design different interfaces, keeping the model the same and changing the interface interactions, and examine how users respond. I could also try keeping the interface the same and changing what features the models use to make predictions, seeing which model functions better for users.

Figure 7
Model Card

- Model Card**
- **Model Details**
 - Information about the stakeholders developing the model.
 - Type: sequential recommender and multi-modal classifier
 - The sequential recommender will be trained to generate lesson plans based on temporal information, lesson information such as its difficulty level and topic, and user information such as their learning level, lesson completion history, and feedback. The classifier will be trained on textual and image data. The models will be examined for personalization fairness and misclassification, aggregation, and evaluation bias.
 - **Intended Use**
 - Primarily intended for individuals with ASD.
 - May be used by anyone who wants to learn more about recognizing emotions and communicating better.
 - May also be used by anyone who wants a classifier to recognize emotions for them in-text, on-screen, or in-person.
 - **Metrics**

- If the ratio of the text classification to image classification usage goes above 70%, we will conduct user studies to investigate the difference between their experiences with the two classifiers.
- If users complete less than 90% of the lessons they started, we will re-evaluate the model's predictions.
- If users press thumbs-down on a recommendation more than 15% of the time, we will reach out to them in-app to solicit written feedback.
- If 60% of users do not return to the app at least once a week, we will conduct user studies to examine the usability of the app.
- **Training and Evaluation data**
 - The dataset for the emotion classifier will be pre-existing and pre-processed to ensure high annotation agreement and labels with a representative distribution.
 - A dataset for the lesson plan generator would be procured to create a curriculum that is helpful for people across the autism spectrum. It would be done by consulting research studies on helpful teachings, and pre-processed by sorting lessons chronologically, labeling their difficulty and topic, and categorizing user history by completion status, performance, and feedback rating.
- **Quantitative Analyses**
 - The lesson plan recommender will be optimized for precision, and the emotion classifier will be optimized for recall.
 - Both systems will also undergo fairness tests.
- **Ethical Considerations**
 - The user's privacy and security will be maintained by keeping their identity and customizations anonymous outside of their personal device and not storing them in any databases. Any user-specific information would only be available at their specific instance of the model and would be inaccessible to data collectors. The only data that would be collected would be features such as frequency of usage, and would not include anything the user entered.
 - Using the technology in an invasive way is a breach of human safety and can warrant legal action. Thus, the app has safeguards to protect the people on whom the classifier may be used and protect stakeholders from lawsuits.
- **Caveats and Recommendations**
 - When using the emotion classifier on people in real-time, it is the user's responsibility to obtain consent. Otherwise, they will violate human safety and privacy and risk legal measures.

Figure 8
UI Sketch



Section 8: Maintain

I believe the models' performances will improve over time. The lesson plan generator's personalization will improve because it is integrating implicit and explicit feedback from the user. Therefore, it will have a better idea of the user's learning level, what types of lessons they want, and when to present different topics to them. I do not

anticipate concept drift in the case when the user is consistently interacting with the tool. However, if a user returns to it after a period of absence, their preferences may have changed from what the model previously learned. Their learning level could have decreased due to a lack of practice, or improved due to the opposite, and what topics they want to visit with the tool may have changed too. Similarly, the emotion classifier's performance will improve over time as it receives feedback about how it presents its responses. Despite emotions remaining relatively constant, how people represent emotions could change, causing me to anticipate concept drift. For instance, the introduction of new slang or cultural shifts in what mood a facial expression indicates would result in changes that the model was not trained on.

I will be versioning the models over time by storing each trained model's weights and metadata, keeping previous versions accessible, and tracking which model version produced certain outputs. I will do this in order to maintain access to previous versions, trace back issues to specific ones, and retain transparency. The models might fail from a security perspective by releasing user information or tying user input to identifiable information. For instance, the lesson plan generator might retain a user's communicative ability, or the classifier might retain information about an uploaded photo. To monitor and mitigate these failures, I will watch out for prompt injections, clean user input, apply guardrails on the models so they don't release information, and bar against unauthorized access to model endpoints.

There are a couple of errors I anticipate occurring. Firstly, a user error I anticipate is the user uploading photographs of people for emotion recognition without their consent. To mitigate this social and legal risk, I will have a screen asking users if they received consent from subjects before uploading photos, to which they have to respond affirmatively to proceed with the tool. Secondly, system errors may include users wanting harder lessons when the model predicted they were not ready for them, blaming it for a lack of personalization, or taking noisy, ambiguous photographs and blaming it for not being able to produce an accurate reading. To mitigate these for the lesson plan generator, I could show the user what level the model actually thinks they are at or a timeline that the model is following. For the classifier, I could add warnings on screen, prompting the user to take in-focus photos, include fewer faces, or tell them that the model is struggling due to ambiguity. Thirdly, context errors could include the model generating a lesson plan that the user was not in the mood for or predicting an emotion that was not present. These specifically occur because the model lacks context and can thus be mitigated by asking for clarification, incorporating situational cues, and presenting the user with a confidence score. I received a peer response that another solution could be for the user to provide a text summary of what has occurred so far, allowing the AI to distinguish the tone of the situation, e.g., humorous, serious, or other. Finally, background errors can happen due to the model making incorrect assumptions about the information the user retained, the cultural context for emotion recognition

tasks, or the camera quality of the input. To mitigate these situations when neither the user nor the system identifies the error, they can be settled through user inputs or training the model to work with more diverse data.

The model is most likely to fail in situations involving ambiguous data, changes in user needs, or mismatched context. To help users recover smoothly, I will provide messages explaining why the system is uncertain, suggest corrections, and offer overrides so users can adjust difficulty or request a different lesson format.

I will be querying users for feedback in order to improve the model or UI, ensure it is helpful for them and works as intended, and identify if they need support with the interface. Implicit feedback will include how often they open the app, what features they favor, and how long they interact with it. This information will allow me to understand their preferences and where there is room for improvement. Explicit feedback will take the form of thumbs up/down for the lessons and classifications, as well as areas for users to input textual responses or complaints. I will incentivize explicit feedback by informing them that it will improve their own experience with the app and align with their needs. The feedback has positive implications, providing more information about how well the model and UI work for users. However, it is important to be wary of biased users who will then exacerbate bias within the models that are retrained on this feedback.

I received a peer response that it will be difficult to obtain feedback on the classifier because the user does not know the correct classification themselves, thus not giving the model a ground truth signal. To clarify, the classifier will not change its predictions based on user feedback. That is, the feedback would not introduce unreliable labels. Instead, it would be used to calibrate how the system presents its outputs, or find out specifically what it is the user likes or dislikes. I also received a peer response that it may be beneficial to also construct a participatory design session with users and their loved ones to evaluate how their emotional perception is evolving as they use the system. This session would indeed help provide an understanding of the tool's impact and usefulness. Inspired by this suggestion, I would also integrate the System Usability Scale to receive more structured feedback about the system.

Section 9: Conclusion/Impact

The product improves long-term understanding of emotion recognition through personalized lesson plans, provides practice for communication skills so that it is easier for users to ask for clarification in communicative contexts, and includes multi-modal classifiers so that users can receive emotion recognition assistance in real-time. Therefore, the main stakeholders, which are individuals with ASD and the people they interact with, will be able to communicate more smoothly with less negative feelings and

have a greater quality of life. The ethical concerns and limitations of this solution, and how they are addressed, is discussed with the following always-on topics.

The system needs to maintain the privacy and security of users, such that their identity and customizations remain anonymous outside of their personal device and are not stored in any databases. To do so, any user-specific information would only be available at their specific instance of the model and would be inaccessible to data collectors. The only data that would be collected would be features such as frequency of usage, and would not include anything the user entered.

Using the technology invasively can warrant being sued, a legal risk, and a breach of human safety. Thus, the system includes safeguards to protect the people on whom the classifier may be used and protect stakeholders from lawsuits. Lawsuits also garner a financial risk for stakeholders, further motivating this safeguard. There is also a psychological human safety risk for users who may become frustrated at misclassifications or poor lesson scores. To prevent these, the system includes suggestions when getting low prediction confidence, such as by asking the user to provide more context or adjust their camera angle, and has motivating messages throughout lessons.

There is a societal risk for users, such that they face negativity in public for using the tool. To protect them against discrimination, the tool has a subtle UI that can disguise its purpose, and takes text input if it is not plausible to obtain photos. Additionally, there is a financial risk for stakeholders in investing in three different AI models, each requiring its own training and maintenance. In this case, the different models are necessary to keep the tool adaptable to a multitude of use cases, but can be adjusted if user feedback does not deem them useful.

The tool does not have any direct governance or political risks. If policies regarding the use of AI evolve, then the tool must be adjusted accordingly. Otherwise, it does not affect greater institutions and follows current regulations. Finally, for the sake of sustainability, the only deep learning models that the system uses are pretrained classifiers. The classifiers must be able to decipher rich representations to have a helpful accuracy for the user, which is why it must take a deep learning form. However, I do not need to create a model from scratch as there are already well-performing emotion recognition models available, thus maintaining sustainability.

Section 10: Final Reflection

Prior to this class, I was taught about the importance of topics such as maintaining transparency, fairness, and security. While I knew these topics were important to consider, they remained vague and abstract in my head until this class concretely asked me how I would implement them and what my reasoning would be.

Through this practice, my opinion changed from thinking that many aspects of a system are just given, e.g., systems are innately non-interpretable and can't help but perpetuate bias. Rather, the most important lesson I gained from this class is how much agency software developers have over an AI tool and what we can tangibly do to put humans first in creating them. I now understand what it means to center humans when approaching new AI solutions, and how to do so.

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AI Discloser: I used ChatGPT to verify my understanding of the reward function, always-on topics, and anticipatory errors, and brainstorm questions for domain experts and ideas for addressing bias in the system.